



BD71, Big Data Spring 26 : Quick Start

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Introduction to Big Data & Hadoop Map-Reduce

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Abstract

This tutored work introduces the core concepts of Big Data and the Hadoop Ecosystem. It is divided into two parts:

1. **Part 1: Theory Assessment.** A multiple-choice section testing your understanding of the 5 V's of Big Data, scaling strategies, and the architecture of the Hadoop Distributed File System (HDFS).
2. **Part 2: Practical Exercises.** You will formulate fundamental HDFS command-line operations and write real Python scripts for MapReduce (Hadoop Streaming protocol) to solve analytical problems using different reduction strategies (summation, maximum, and counting).

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1 Part 1: Multiple Choice Questions (Theory & HDFS)

Select the best answer for each of the following questions based on the lecture material.

1. Which of the “5 V’s of Big Data” refers specifically to the quality, trustworthiness, and reliability of the data being analyzed?
 - a) Volume
 - b) Velocity
 - c) Variety
 - d) Veracity
2. According to Gartner, approximately what percentage of enterprise data is considered unstructured (e.g., text, images, videos)?
 - a) 20%
 - b) 50%
 - c) 80%
 - d) 95%
3. When dealing with increasing data demands, what does the strategy of “Scaling Out” entail?
 - a) Upgrading the CPU and RAM of a single existing server.
 - b) Adding new equipment (servers/nodes) and distributing workloads across them.
 - c) Transitioning from a distributed file system to a monolithic relational database.
 - d) Migrating structured data into an unstructured format.
4. Introduced in Hadoop 2.0, which component is responsible for separating resource management from job scheduling, thereby allowing multiple processing engines to run simultaneously?
 - a) HDFS
 - b) MapReduce
 - c) ZooKeeper
 - d) YARN
5. Which characteristic of Big Data describes the speed at which data is generated, processed, and analyzed?
 - a) Volume
 - b) Value
 - c) Velocity
 - d) Variety
6. Based on the Big Data Model, which of the following is considered an example of **structured** data?
 - a) Relational databases and CSV files
 - b) Social media posts and emails

- c) Audio and video files
 - d) Word processing documents
7. In the MapReduce programming model, what is the primary function of the **Reduce** phase?
- a) To divide the input data into smaller, manageable chunks for processing.
 - b) To process the raw input data and generate intermediate key-value pairs.
 - c) To allocate computational resources across the Hadoop cluster.
 - d) To aggregate the intermediate data output from the map phase and generate the final output.
8. Within the YARN architecture, which component runs on worker machines and is responsible for monitoring their resource usage (CPU, memory, disk) and reporting it to the Resource Manager?
- a) Application Master
 - b) NodeManager
 - c) Container
 - d) Scheduler
9. What is the typical block size used in a Distributed File System like HDFS to store data chunks?
- a) 4 KB – 32 KB
 - b) 1 MB – 5 MB
 - c) 64 MB – 128 MB
 - d) 1 GB – 2 GB
10. Which component of the HDFS architecture is strictly responsible for managing the file system's metadata and the directory structure?
- a) DataNode
 - b) NameNode
 - c) Secondary NameNode
 - d) NodeManager
11. How does HDFS fundamentally ensure fault tolerance when a node fails in the cluster?
- a) The Secondary NameNode takes over immediately.
 - b) The failed DataNode is automatically repaired by YARN.
 - c) Files are divided into chunks and replicated across multiple DataNodes.
 - d) The NameNode retrieves the lost data from the local filesystem.
12. What is the correct HDFS command to upload a file named `data.txt` from the local Linux directory to HDFS?
- a) `hadoop fs -get data.txt`
 - b) `hadoop fs -put data.txt`
 - c) `hadoop fs -mv data.txt`

- d) `hadoop fs -upload data.txt`
13. Which of the following HDFS commands is used to download a file from Hadoop back to your local disk?
- a) `hadoop fs -put`
 - b) `hadoop fs -cat`
 - c) `hadoop fs -get`
 - d) `hadoop fs -fetch`
14. When a client wants to read a file from HDFS, which component does it contact **first** to find the IP addresses of the required data blocks?
- a) The Core Switch
 - b) The DataNode
 - c) The Secondary NameNode
 - d) The NameNode
15. In HDFS, what does the Secondary NameNode primarily download from the primary NameNode to perform its maintenance and checkpointing tasks?
- a) Data blocks and replicas
 - b) FSImage and Edit Logs
 - c) Mapper and Reducer scripts
 - d) The `core-site.xml` configuration file
16. Which HDFS command allows you to view the **last few lines** of a file directly from the distributed file system without downloading the entire file?
- a) `hadoop fs -cat file.txt`
 - b) `hadoop fs -tail file.txt`
 - c) `hadoop fs -less file.txt`
 - d) `hadoop fs -end file.txt`
17. In which Hadoop configuration file would you typically define the HDFS replication factor (`dfs.replication`) and the NameNode/DataNode directory paths?
- a) `core-site.xml`
 - b) `mapred-site.xml`
 - c) `yarn-site.xml`
 - d) `hdfs-site.xml`
18. According to the HDFS write pipeline, what happens immediately after the HDFS Client successfully transfers the data blocks to the required DataNodes?
- a) The client updates the metadata directly.
 - b) The NameNode formats the HDFS.
 - c) The DataNodes send an acknowledgment (ACK) back up the pipeline.

- d) The Secondary NameNode creates a checkpoint.
19. In HDFS, which component is responsible for storing the actual data blocks and handling read and write requests from the file system's clients?
- a) NameNode
 - b) Secondary NameNode
 - c) DataNode
 - d) ApplicationMaster
20. In the Hadoop Distributed File System (HDFS) architecture, is the Secondary NameNode a direct backup that takes over if the primary NameNode crashes?
- a) Yes, it acts as a hot standby.
 - b) No, it only negotiates resources for MapReduce.
 - c) No, it only periodically consolidates metadata.
 - d) Yes, but only for block storage, not metadata.

2 Part 2: Practical Exercises

2.1 Exercise 1: Basic HDFS Commands

Assume you have access to a terminal configured with Hadoop. Write the exact command-line instructions required to perform the following sequence of operations:

1. Create a new directory named `lab_data` in your HDFS home directory.
2. Create a simple local text file named `my_data.txt` (e.g., using the `echo` command) and upload it to the `lab_data` directory you just created in HDFS.
3. Display the contents of the uploaded file directly from HDFS to verify it without downloading it.
4. Download the file from HDFS back to your local machine, saving it as `my_data_copy.txt`.

2.2 Exercise 2: MapReduce Python (Summation Reduction)

You are provided with a dataset of store purchases (`purchases.txt`). Each line represents a transaction and is formatted with 6 fields separated by a tab (`\t`):

```
Date \t Time \t Store \t Product Category \t Cost \t Payment Method
```

Write the `mapper.py` and `reducer.py` scripts to compute the **total sales amount for each Payment Method** (e.g., Visa, MasterCard, Cash).

1. Write the `mapper.py` code. Extract the correct Key and Value.
2. Write the `reducer.py` code to sum the costs per payment method.

2.3 Exercise 3: MapReduce Python (Maximum Reduction)

Using the exact same `purchases.txt` dataset structure as in Exercise 2, we now want to find the **highest individual sale amount** recorded for each **Product Category** (e.g., Toys, Consumer Electronics).

1. Write the corresponding `mapper.py` code.
2. Write the corresponding `reducer.py` code to evaluate and output the maximum cost per category instead of the sum.

2.4 Exercise 4: MapReduce Python (Counting Reduction)

You are analyzing an Apache web server log file (`access_log`). Each entry has the following format, separated by spaces:

```
10.223.157.186 - - [15/Jul/2009:15:50:35 -0700] "GET /assets/js/lowpro.js HTTP/1.1" 200 10469
```

The first field (`10.223.157.186`) represents the IP address of the client making the request.

Write the `mapper.py` and `reducer.py` scripts to calculate the **total number of site accesses per IP address** (i.e., how many times each IP appears in the log).